Garlock
SEALING TECHNOLOGIES®
an EnPro Industries company

# Garlock 5500

## MATERIAL PROPERTIES*

| | |
|---|---|
| **Color:** | Gray |
| **Composition:** | Inorganic fibers with a nitrile binder |
| **Fluid Services**[1]**:** | Saturated steam[3], most refrigerants, water, oils, gasoline and aliphatic hydrocarbons |
| **Temperature**[2], °F (°C) | |
| Minimum: | -100 (-73) |
| Continuous Max: | +550 (+288) |
| Maximum: | +800 (+427) |
| **Pressure**[2], Maximum, psig (bar): | 1200 (83) |
| **P x T (max.)**[2], psig x °F (bar x °C) | |
| 1/32 and 1/16": | 400,000 (14,000) |
| 1/8": | 275,000 (9,600) |
| **Meets Specification:** | ABS (American Bureau of Shipping) and Fire Safe |

## PHYSICAL PROPERTIES*

| | | | | |
|---|---|---|---|---|
| **ASTM F36** | **Compressibility**, range, %: | 7-17 | | |
| **ASTM F36** | **Recovery**, %: | 50 | | |
| **ASTM F38** | **Creep Relaxation**, %: | 15 | | |
| **ASTM F152** | **Tensile**, Across Grain, psi (N/mm$^2$): | 1500 (10) | | |
| **ASTM F1315** | **Density**, lbs./ft.$^3$ (grams/cm$^3$): | 110 (1.76) | | |
| **ASTM F433** | **Thermal Conductivity (K)**, W/m°K (Btu.·in./hr.·ft.$^2$·°F): | 0.43-0.53 (3.00-3.65) | | |
| **ASTM D149** | **Dielectric Properties**, range, volts/mil. | | | |
| | Sample conditioning | 1/16" | 1/8" | |
| | 3 hours at 250°F: | 284 | 245 | |
| | 96 hours at 100% Relative Humidity: | - | - | |
| **ASTM F586** | **Design Factors** | 1/16" | 1/8" | |
| | "m" factor: | 6.6 | 6.6 | |
| | "y" factor, psi (N/mm$^2$): | 2600 (17.9) | 3300 (22.8) | |
| **ROTT** | **Gasket Constants**, 1/16": | Gb=1,247 | a=0.249 | Gs=11.0 |
| **ASTM F104** | **Line Call Out**: | F712103A9B4E23K7L501M4[(4)] | | |

## SEALING CHARACTERISTICS*

| | ASTM F37B Fuel A | ASTM F37B Nitrogen | DIN 3535- 4 Gas Permeability |
|---|---|---|---|
| **Gasket Load**, psi (N/mm2): | 500 (3.5) | 3000 (20.7) | 4640 (32) |
| **Internal Pressure**, psig (bar): | 9.8 (0.7) | 30 (2) | 580 (40) |
| **Leakage** | **0.2 ml/hr.** | **1.0 ml/hr.** | **0.05 cc/min** |

## IMMERSION PROPERTIES* - ASTM F146 Fluid Resistance after Five Hours

| | ASTM #1 Oil 300°F (150°C) | ASTM IRM #903 300°F (150°C) | ASTM Fuel A 70-85°F (20-30°C) | ASTM Fuel B 70-85°F (20-30°C) |
|---|---|---|---|---|
| **Thickness Increase**, (%) | 0-10 | 0-15 | 0-10 | 0-15 |
| **Weight Increase,** (%) | <15 | - | <10 | <15 |
| **Tensile Loss,** (%) | - | <40 | - | - |

**Notes:**
This is a general guide and should not be the sole means of selecting or rejecting this material. ASTM test results in accordance with ASTM F-104; properties based on 1/32" (0.8mm) sheet thickness unless otherwise mentioned.

* Values do not constitute specification Limits

1 See Garlock chemical resistance guide.

2 Based on ANSI RF flanges at our preferred torque. When approaching maximum pressure, continuous operating temperature, minimum temperature or 50% of maximum PxT, consult Garlock Applications Engineering. Minimum temperature rating is conservative.

3 Minimum recommended assembly stress = 4,800psi. Preferred assembly stress = 6,000-10,000psi. Gasket thickness of 1/16" strongly preferred. Retorque the bolts/studs prior to pressurizing the assembly. For saturated steam above 150psig or superheated steam, consult Garlock Engineering.

4 A9: Leakage in Fuel A (Isooctane), Gasket Load = 500psi (3.5N/mm2), Pressure = 9.8psig (0.7bar): Typical = 0.2ml/hr, Max = 1.0ml/hr. A9: Leakage in Nitrogen, Gasket Load = 3,000psi (20.7N/mm2), Pressure = 30psig (2bar): Typical = 0.5ml/hr, Max = 1.5ml/hr.